JXC-5010: A Low-Cost 3D-Printed Cycloidal Actuator from a Drone Motor — Design and Build Plan

JX1 Humanoid Project

***Abstract*—We design a 3D-printable joint actuator around the low-cost parts initially proposed: a 5010 360 KV drone motor and a SimpleFOCmini driver. Analysis shows the driver's 2.5 A limit caps output torque at about 1.0 N·m regardless of the gearbox, so we replace it with the ST B-G431B-ESC1 (40 A peak, CAN, about US$19), which lifts the motor-limited torque to 7.8 N·m. Comparing cycloidal, strain-wave and planetary reducers for fused-filament printing, we select a single-stage, dual-disc cycloidal reducer of 20:1 with 21 steel ring pins and rolling output pins. Stress checks show that the printed discs, not the motor, limit the actuator to about 10 N·m peak and 4 N·m continuous, so a costlier motor adds little at this size. The resulting JXC-5010 (Ø82×64 mm, about 0.46 kg) delivers about 8 N·m peak, 4 N·m continuous and 45 rad/s no-load at 24 V for about INR 9,993 in parts. We provide printable STL/DXF files, a bill of materials, print settings, an assembly sequence, firmware steps and a test plan.**

***Index Terms*—Cycloidal drive, 3D printing, quasi-direct drive, BLDC, field-oriented control, low-cost robotics.**

I. Goal and Starting Point

THE goal is the highest-torque joint actuator that can be prototyped by fused-filament (FDM) printing at roughly the cost of the parts initially chosen: a generic 5010 360 KV outrunner (INR 2,239) and a SimpleFOCmini driver (INR 729). The design follows the open-source Berkeley Humanoid Lite (BHL) actuators, which walk a 16 kg humanoid on printed 15:1 cycloidal gearboxes driven by 5010 and 6512 drone motors [1].

II. Component Selection

A. The driver is the bottleneck

A motor's torque constant can be estimated from its KV rating as $K_t \approx 8.27/K_V$ N·m/A, so the 5010 360 KV gives about 0.023 N·m/A (measure it; Section IX). The output torque of a geared actuator is

$$T_{\text{out}} = K_t \, I \, G \, \eta, \tag{1}$$

with current I, ratio G and gearbox efficiency $\eta \approx 0.85$. The SimpleFOCmini (DRV8313) delivers only 2.5 A, so even at 20:1 the actuator reaches about 1.0 N·m. The motor itself is rated for 20–40 A drone ESCs; a driver supplying 20 A multiplies torque by eight. Table I compares options. The **ST B-G431B-ESC1** is chosen: 40 A peak with forced air (20 A without), up to 6S, an STM32G431 with CAN on board, a USB ST-Link for programming, support in the SimpleFOC library and in BHL's "Recoil" CAN firmware [1], [2], at about US$19. It is often out of stock in India (Evelta listed it at INR 2,791); order from ST's eStore or a global distributor. The Flipsky ODESC V4.2 (INR 7,350 on Amazon.in) is the in-stock fallback. The SimpleFOCmini remains useful for learning FOC on the bench or for very light joints with a gimbal motor.

B. The motor

Keep the 5010 360 KV. Its high KV is matched by a high ratio (20:1 still gives 45 rad/s at the joint). A low-KV 5010 such as the Tarot 130 KV (INR 4,779) doubles the motor-limited torque at lower current, but Section IV shows the printed gearbox caps peak torque near 10 N·m, so the upgrade mainly buys cooler running. Spend the money on the driver first.

C. Which gearbox for a printed prototype

Table II compares the three reducer types for FDM printing. A strain-wave (harmonic) drive needs a thin, repeatedly flexed spline: printed in plastic it has low torque and fatigues quickly. Printed planetaries measured 60–80 % efficiency and 0.6–2.6° backlash [3], and a single stage reaches only about 6–10:1, too little for a 360 KV motor. A cycloidal reducer reaches 10–25:1 in one stage, spreads load over many lobes at once, uses steel pins for every sliding contact, tolerates shock, and has been validated in print: BHL measured about 90 % efficiency, 1.3° backlash and a 60 h life test [1]. **We choose a cycloidal reducer.**

III. Reducer Design

A. Geometry

With N ring pins on a circle of radius R, pin radius r (plus a fit clearance) and eccentricity e, the disc profile is

$$\begin{aligned} x(t) &= R\cos t - r\cos(t+\psi) - e\cos(Nt), \\ y(t) &= -R\sin t + r\sin(t+\psi) + e\sin(Nt), \\ \psi &= \operatorname{atan2}\!\big(\sin((1-N)t),\ \tfrac{R}{eN} - \cos((1-N)t)\big), \end{aligned} \tag{2}$$

Prepared October 2026. Design script, print files and this plan are in `actuators/diy_cycloid/` of the `thenar-jx1` repository. All performance values are calculated and must be confirmed on the bench (Section IX).

TABLE I
MOTOR AND DRIVER OPTIONS WITH THE 20:1 PRINTED CYCLOID AT 24 V (CALCULATED; † = INITIALLY CHOSEN PART)

Motor	Driver	K_t [N·m/A]	I_{peak} [A]	Peak [N·m]	Cont. [N·m]	No-load [rad/s]	Motor + driver [INR]
5010 360KV†	SimpleFOCmini†	0.023	2.5	1.0	0.8	45	2,968
5010 360KV†	ST B-G431B-ESC1	0.023	20.0	7.8	3.9	45	5,039
5010 360KV†	Flipsky ODESC V4.2	0.023	20.0	7.8	3.9	45	9,589
Tarot 5010 130KV	SimpleFOCmini†	0.064	2.5	2.7	2.2	16	5,508
Tarot 5010 130KV	ST B-G431B-ESC1	0.064	15.0	10.0*	4.0	16	7,579
Tarot 5010 130KV	Flipsky ODESC V4.2	0.064	15.0	10.0*	4.0	16	12,129

*Limited by the printed gearbox, not the motor. Continuous torque is also capped at the printed-gear limit.

TABLE II
REDUCER TYPES FOR A 3D-PRINTED PROTOTYPE

Type	For printing	Against
Cycloidal	10–25:1 in one stage; many lobes share load; steel pins take contact; shock tolerant; proven in PLA (BHL)	eccentric vibration (use two discs at 180°); less back-drivable above 20:1
Strain-wave	very low backlash	printed flexspline is weak and fatigues; hard to make
Planetary	back-drivable, simple	60–80 % efficient printed; small teeth wear; ≤10:1 per stage

which has $N-1$ lobes and gives a reduction $G = N-1$ with the output taken from the disc through pins in oversized holes of diameter $d_{\text{roller}} + 2e$. The short-amplitude coefficient $\lambda = eN/R$ must be below 1 to avoid undercut. The chosen geometry is $N = 21$ ($G = 20$), $R = 32$ mm, Ø4 mm pins, $e = 1.0$ mm, $\lambda = 0.66$, two 7 mm discs at 180° on a double eccentric, 6 output pins on $R_w = 21$ mm with MR74ZZ rollers, and a 0.1 mm profile clearance for FDM tolerance (Fig. 1). The disc spans radius 28.9–30.9 mm, the pin pitch is 9.6 mm, the smallest concave profile radius 3.3 mm exceeds the pin radius, and the walls around the Ø9.0 mm output holes are 3.5 mm (to the bearing bore) and 3.4 mm (to the lobe root).

B. Loads and stresses

For a dual-disc drive each disc is assumed to carry 55 % of the output torque T_c. The largest ring-pin and output-pin forces are

$$F_{\text{pin}} = \frac{4T_c}{e\,N\,(N-1)}, \qquad F_{\text{out}} = \frac{4T_c}{Z_w R_w}, \tag{3}$$

and the contact pressure follows from Hertz line contact, $p = \sqrt{FE^*/(\pi b R_{\text{eff}})}$, with $E^* \approx 4$ GPa for PLA/PETG and 6 GPa for PA-CF on steel. Table III gives the results. Printed PLA yields at about 60–70 MPa in compression, so the printed discs set a peak limit of about 10 N·m; for long life the continuous torque should stay near 4 N·m (contact pressure $\lesssim$45 MPa). The hardened Ø4 output pins see 128–320 MPa in bending, well within their strength.

IV. MECHANICAL LAYOUT

Fig. 2 shows the stack. The motor base bolts to an adapter plate; a spacer ring surrounds the motor; the eccentric cam bolts to the rotor bell's propeller-mount holes and is centred on the 4 mm shaft; the ring-pin housing holds 21 pressed Ø4×20 dowels half-exposed in grooves; the carrier, with six Ø4 pins pressed through it, runs in a 6810 bearing held by a retainer ring; six M3 bolts clamp the stack. The two parts marked "MEASURE_FIRST" assume a 25 mm motor-base bolt circle and a 16 mm propeller-mount circle: measure your motor and change the two parameters in `cycloid_actuator.py` before printing. Fig. 3 shows the printed parts.

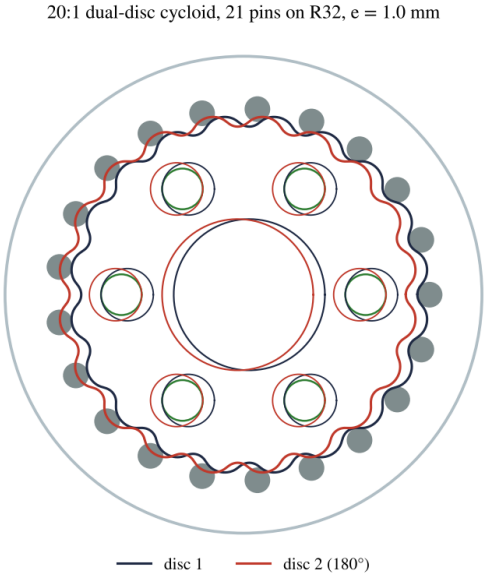


Fig. 1. Cycloidal stage: two discs 180° apart, 21 ring pins and 6 output rollers (green).

TABLE III
REDUCER CHECKS AT THREE OUTPUT TORQUES (CALCULATED)

Quantity	4 N·m	8 N·m	10 N·m
Torque per disc [N·m]	2.2	4.4	5.5
Max ring-pin force [N]	21	42	52
Max output-pin force [N]	70	140	175
Ring-pin contact, PLA/PETG [MPa]	44	62	69
Ring-pin contact, PA-CF [MPa]	53	76	85
Output-roller contact, PLA/PETG [MPa]	34	48	53
Output-pin bending [MPa]	128	256	320

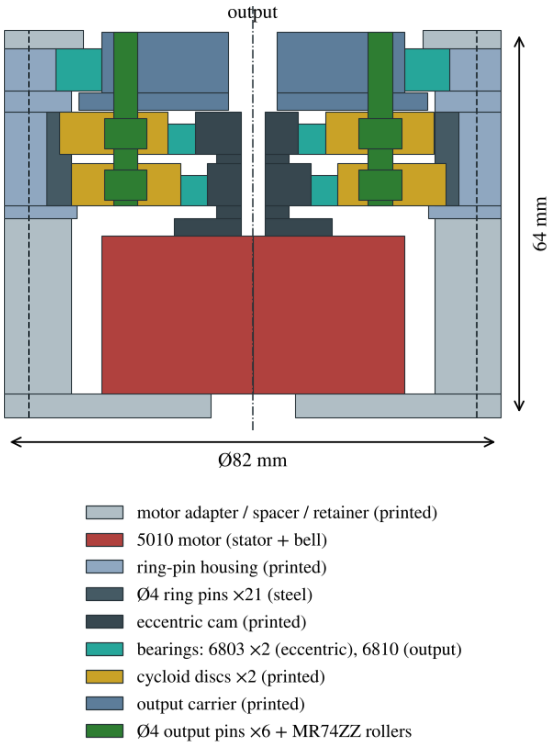


Fig. 2. JXC-5010 section to scale (Ø82×64 mm).

V. Bill of Materials

Table IV lists the parts, about INR 9,993 including a bench supply. Prices marked E are estimates to be confirmed.

VI. Printing

- *Test coupon first*: print a 20 mm block with Ø3.95/4.0/4.05 holes and a Ø26 bore; pick the hole size that presses a dowel in firmly and the bore that holds a 6803 by hand press. Adjust `d_pin` offsets and `clearance` in the script.
- *v0 (fit and function)*: PLA+, 0.2 mm layers, 4–5 walls. Discs flat on the bed, 100 % infill (they carry the contact load); housing output-side up, 40 % gyroid; carrier and cam 80–100 %.
- *v1 (durable)*: PETG-CF or PA-CF for discs and cam (higher temperature and stiffness; PA-CF needs a hardened nozzle and dry filament). Keep PLA away from the motor, which can exceed PLA's 55 °C softening point.
- Disable elephant-foot on the discs (or chamfer 0.3 mm); remove all stringing in the holes; dry-fit before greasing.

VII. Assembly

1) Press the 21 ring pins into the housing (vise, parallel jaws); check that all protrude equally.
2) Press a 6803 bearing into each disc; press the six Ø4 pins through the carrier; slide two MR74ZZ rollers per disc onto each pin.
3) Bolt the motor to the adapter plate; fit the spacer ring. Glue the diametric magnet to the motor's rear shaft end; mount the encoder 1–2 mm below it on a printed bracket.
4) Bolt the eccentric cam to the rotor bell; check run-out by eye while spinning slowly.
5) Grease the pins. Fit the housing; drop disc 1 onto the lower journal, then disc 2 rotated 180° onto the upper journal, rotating the motor slowly until each disc meshes.
6) Insert the carrier pins through both discs; press the 6810 bearing over the carrier hub into the housing seat; fit the retainer and the six M3 through-bolts. Turn the output by hand: it should rotate the motor 20 times per turn without binding.

VIII. Electronics and Firmware

Power the B-G431B-ESC1 from a current-limited 24 V supply (start at 3 A). Wire the motor phases and the MT6701 (ABZ or SSI) or AS5600 (I^2C) encoder. Program it over its USB ST-Link with the Arduino IDE (STM32duino core) and the SimpleFOC library: (1) open-loop velocity to confirm the phases; (2) sensor alignment and pole-pair check (7 pole pairs for a 12N14P motor); (3) torque (current) mode with a 3 A limit; (4) velocity and position loops; (5) raise the current limit in steps to 20 A while watching temperature. For a robot, move to BHL's Recoil firmware for CAN control and add an output-side encoder or homing switch, since a 20:1 stage makes the motor encoder ambiguous after power-up.

IX. Test Plan and Expected Results

- K_t: lock the output with a 100 mm lever on a kitchen scale; step current to 2, 4, 8 A; $K_t = T/(I\,G\,\eta)$. Update the script.
- *Peak torque*: step to 20 A for 2 s; expect ≈7.8 N·m. Stop if the discs creak or the scale reading drops.
- *Backlash*: lock the motor, rock the output with a 200 mm arm and a dial gauge (BHL: 1.3° max).
- *Thermal*: 30 min at 3–4 N·m; motor below 80 °C, housing below 50 °C (PLA).
- *Life*: replay a sinusoidal 0–4 N·m load for 20 h; recheck backlash and efficiency every 4 h.

Expected: about 7.8 N·m peak (2–3 s), 3.9 N·m continuous, 45 rad/s no-load at 24 V, about 0.46 kg.

X. Upgrade Path

Beyond about 10 N·m the printed discs are the limit. Next steps in order of cost: print discs in PA-CF; have the discs laser-cut from 3 mm steel or 6 mm aluminium (the DXF in `cad/` is ready for a laser shop); scale the stage to $R \approx 40$ mm with a 6012 or 8108-class motor for 20–30 N·m, the size of JX1's ankle and hip-yaw joints; or move to the JXA family of the companion paper [6].

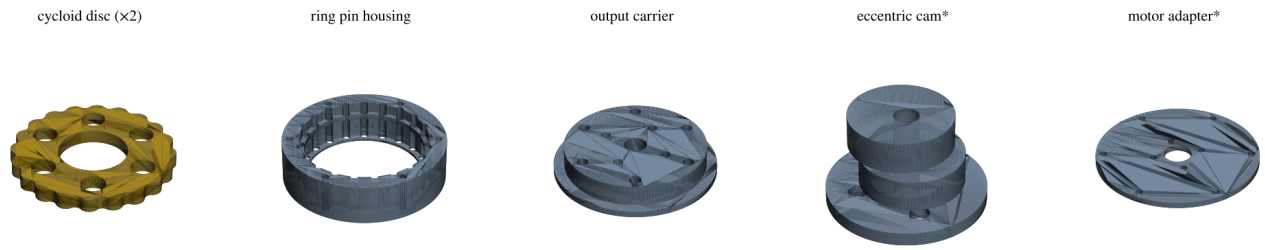


Fig. 3. Printed parts generated by the design script (STL and DXF in `cad/`); spacer and retainer rings not shown.

TABLE IV
Bill of materials (INR incl. GST; E = estimate)

Item	Part	Qty	Unit	Total	Source / note
Motor	5010 360KV outrunner (Robu SKU 45571)	1	2,239	2,239	your pick; keep it
Driver	ST B-G431B-ESC1 (STM32G431, 40 A peak, CAN, USB ST-Link)	1	2,800	2,800	E: US$18.59 at ST eStore + shipping/duty
Encoder	MT6701 14-bit magnetic encoder module	1	760	760	E: Ubuy 2 for INR 1,525; AS5600 (12-bit) is the budget fallback
Encoder magnet	6×2.5 mm diametric NdFeB	2	40	80	E
Output bearing	6810-2RS (50×65×7), FBJ	1	470	470	bearinghouse.in
Eccentric bearings	6803-2RS (17×26×5)	2	100	200	E
Output rollers	MR74ZZ (4×7×2.5), 2 per pin per disc	24	30	720	E
Ring pins	Ø4×20 steel dowel, pack of 100 (use 21)	1	624	624	IndustryBuying (CE)
Output pins	Ø4×30 dowel or Ø4 silver-steel rod cut to 29 mm	6	25	150	E
Fasteners	6× M3×65 + nuts, M3×8, M3 heat-set inserts	1	300	300	E
Filament	≈180 g PLA+ (v0) / PETG-CF or PA-CF (v1)	1	250	250	INR 700–4,800/kg
Grease	lithium or PTFE grease	1	200	200	E
Power	24 V 10 A supply (or a 6S LiPo you have)	1	1,200	1,200	E; current-limited bench supply preferred
Total				**9,993**	

XI. Conclusion

The initially chosen motor is adequate; the driver is not. With a B-G431B-ESC1 and a printed 20:1 dual-disc cycloid, the 5010 360 KV makes an actuator of about 8 N·m peak for roughly the price of a single hobby servo-class joint, limited by the printed discs. All files to print, assemble and test it are provided.

References

[1] Q. Liao *et al.*, "Berkeley Humanoid Lite: An open-source, accessible, and customizable 3D-printed humanoid robot," arXiv:2504.17249, 2025.

[2] A. Skuric, H. S. Bank, R. Unger, O. Williams, and D. González-Reyes, "SimpleFOC: A field oriented control (FOC) library for controlling brushless direct current (BLDC) and stepper motors," *J. Open Source Softw.*, vol. 7, no. 74, p. 4232, 2022.

[3] COMPAct: open-source printed planetary actuators (IISc / ARTPARK), summarised in `research/raw/actuator_technology_raw.md`, §3.

[4] J. G. Blanche and D. C. H. Yang, "Cycloid drives with machining tolerances," *J. Mech. Transm. Autom. Des.*, vol. 111, no. 3, pp. 337–344, 1989.

[5] STMicroelectronics, "B-G431B-ESC1 Discovery kit," user manual UM2516.

[6] JX1 Humanoid Project, "Design and techno-economic analysis of in-house quasi-direct-drive actuators for humanoid robots in India," Oct. 2026.